



1. red

$$\tau \dot{x} + x = z$$

Zanimanje:

- odziv za tipične z
- skica
- ali senzor sledi vhodu?

Laplace

$$f(t) \rightarrow F(s)$$

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

$$\tau \dot{x} + x = z / \mathcal{L}$$

$$\tau s \mathcal{L}\{x\} + \mathcal{L}\{x\} = \mathcal{L}\{z\}$$

$$X(\tau s + 1) = Z$$

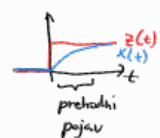
$$X(s) = \frac{1}{\tau s + 1} Z(s) \rightarrow \mathcal{L}^{-1} \rightarrow x(t)$$

$H(s)$ prenosna funkcija

Primer:

$$z(t) = 1$$

$\Delta(t) = z(t) - x(t) \xrightarrow{t \gg \tau} \text{konst.} \Rightarrow \text{sledi}$
 $f(t) \Rightarrow \text{ne sledi}$



$$x(t) = ?$$

$$X(s) = H(s) Z(s) = \frac{1}{1+\tau s} \cdot \frac{1}{s} = \frac{A}{1+\tau s} + \frac{B}{s} = \frac{1}{s} - \frac{\tau}{1+\tau s} = \frac{1}{s} - \frac{1}{s + 1/\tau}$$

$$1 = As + B + B\tau s = B + (A + B\tau)s$$

$$B = 1, A = -1/\tau$$

$$x(t) = 1 - e^{-t/\tau}$$

$\Delta(t) = e^{-t/\tau} \xrightarrow{t \gg \tau} 0$ sledi vhodu brez zamika

$$z(t) = kt$$

$$Z(s) = \frac{k}{s^2}$$

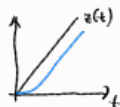
$$X(s) = \frac{1}{1+\tau s} \cdot \frac{k}{s^2} = \left(\frac{A}{1+\tau s} + \frac{Bs+C}{s^2} \right) k$$

$$1 = As^2 + Bs + C + \tau Bs^2 + C\tau s = C + (B + C\tau)s + (A + \tau B)s^2$$

$$C = 1, B = -\tau, A = \tau^2$$

$$X(s) = \left(\frac{\tau}{1/\tau + s} - \frac{\tau}{s} + \frac{1}{s^2} \right) k$$

$$x(t) = k(t - \tau + \tau e^{-t/\tau}) = kt - k\tau(1 - e^{-t/\tau}) \xrightarrow{t \gg \tau} k(t - \tau)$$



$$\Delta(t \gg \tau) = k\tau \text{ sledi vhodu z zamikom}$$

$$z(t) = at^2$$

$$Z(s) = \frac{2a}{s^3}$$

$$X(s) = \frac{1}{1+\tau s} \cdot \frac{2a}{s^3} = 2a \left(\frac{A}{1+\tau s} + \frac{Bs^2 + Cs + D}{s^3} \right) = 2a \left(\frac{1}{s^3} - \frac{\tau}{s^2} + \frac{\tau^2}{s} - \frac{\tau^3}{1+\tau s} \right)$$

$$s^0: 1 = D$$

$$s^1: 0 = C + \tau D \rightarrow C = -\tau$$

$$s^2: 0 = B + \tau C \rightarrow B = \tau^2$$

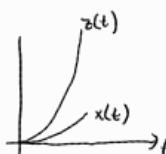
$$s^3: 0 = A + B\tau \rightarrow A = -\tau^3$$

$$x(t) = at^2 - 2a\tau t + 2a\tau^2 - 2a\tau^2 e^{-t/\tau}$$

$$t \gg \tau:$$

$$x(t) = at^2 - 2a\tau t + 2a\tau^2$$

$$\Delta(t) = 2a\tau t - 2a\tau^2 = f(t) \rightarrow \text{ne sledi vhodu}$$



$$\text{I: } z(t) = kt \rightarrow x(t) = kt - k\tau(1 - e^{-t/\tau})$$

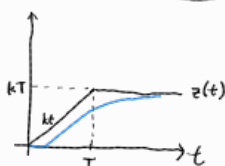
$$\text{II: } z(t) = A \rightarrow x(t) = A(1 - e^{-t/\tau})$$

2. opcija:

$$z(t) = kt - k(t - \tau)\theta(t - \tau)$$

$$Z(s) = \frac{k}{s^2} - \frac{k}{s^2} e^{-s\tau}$$

$$X(s) = \frac{1}{1+\tau s} \cdot \frac{k}{s^2} (1 - e^{-s\tau}) \rightarrow x(t) = kt - k\tau(1 - e^{-t/\tau}) - (k(t - \tau) - k\tau(1 - e^{-(t-\tau)/\tau}))\theta(t - \tau)$$



2. red

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2x = \omega_0^2z (+2\zeta\omega_0\dot{z})$$

$$s^2X + 2\zeta\omega_0sX + \omega_0^2X = \omega_0^2Z (+2\zeta\omega_0sZ)$$

$$\frac{X(s)}{Z(s)} = H(s) = \frac{\omega_0^2(1+2\zeta\omega_0s)}{s^2+2\zeta\omega_0s+\omega_0^2}$$

Primer

$$z(t) = 1$$

$$x(t) = ?$$

$$Z(s) = \frac{1}{s}$$

$$X(s) = \frac{1}{s} \frac{\omega_0^2}{s^2+2\zeta\omega_0s+\omega_0^2} = \omega_0^2 \left(\frac{C}{s} + \frac{As+B}{s^2+2\zeta\omega_0s+\omega_0^2} \right)$$

$$C=1$$

$$A=-1$$

$$B=-2\zeta\omega_0$$

$$X(s) = \frac{1}{s} - \frac{s+2\zeta\omega_0}{s^2+2\zeta\omega_0s+\omega_0^2}$$

$$s^2+2\zeta\omega_0s+\omega_0^2 = (s+\zeta\omega_0)^2 - \zeta^2\omega_0^2 + \omega_0^2$$

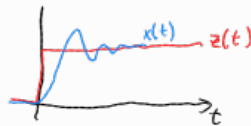
$$\frac{s+2\zeta\omega_0}{(s+\zeta\omega_0)^2 + \omega_0^2(1-\zeta^2)} \rightarrow \alpha = -\zeta\omega_0$$

$$\frac{s+\zeta\omega_0}{(s+\zeta\omega_0)^2 + \omega_0^2(1-\zeta^2)} + \frac{\zeta\omega_0}{-11-} \frac{\sqrt{1-\zeta^2}}{\sqrt{1-\zeta^2}}$$

$$X(s) = \frac{1}{s} - \frac{s+\zeta\omega_0}{-11-} - \frac{\zeta}{11-\zeta^2} \frac{\omega_0\sqrt{1-\zeta^2}}{-11-}$$

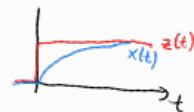
$$x(t) = 1 - \left(\cos(\omega_0\sqrt{1-\zeta^2}t) e^{-\zeta\omega_0 t} + \frac{\zeta}{11-\zeta^2} \sin(\omega_0\sqrt{1-\zeta^2}t) e^{-\zeta\omega_0 t} \right)$$

a) $\zeta < 1 \rightarrow$ podkritično dušenje



b) $\zeta = 1 \rightarrow$ kritično dušenje

$$x(t) = 1 - \left(1 + \zeta\omega_0 t \frac{\sin\omega_0 t}{\omega_0 t} \right) e^{-\zeta\omega_0 t} = 1 - (1 + \omega_0 t) e^{-\omega_0 t}$$



c) $\zeta > 1 \rightarrow$ nadkritično dušenje

$$x(t) = 1 - \left(\cos(i\omega_0\sqrt{\zeta^2-1}t) + \frac{\zeta}{i\sqrt{\zeta^2-1}} \sin(i\omega_0\sqrt{\zeta^2-1}t) \right) e^{-\zeta\omega_0 t}$$

$\rightarrow$ eksponentno približevanje

d) optimalno dušenje

$$\zeta_{opt} = \frac{1}{\sqrt{2}} < 1$$

$$z(t) = kt$$

$$x(t) = ?$$

$$Z(s) = \frac{k}{s^2}$$

$$X(s) = \frac{k}{s^2} \frac{\omega_0^2}{s^2+2\zeta\omega_0s+\omega_0^2} = k\omega_0^2 \left(\frac{As+B}{s^2+2\zeta\omega_0s+\omega_0^2} + \frac{Cs+D}{s^2} \right)$$

$$1 = (A+C)s^3 + (B+D) + 2\zeta\omega_0Cs + (2\zeta\omega_0D+\omega_0^2)s + D\omega_0^2$$

$$D = \frac{1}{\omega_0^2}$$

$$C = -\frac{2\zeta D}{\omega_0} = -\frac{2\zeta}{\omega_0^3}$$

$$B = -D - 2\zeta\omega_0C = -\frac{1}{\omega_0^2} + \frac{4\zeta^2}{\omega_0^2} = \frac{4\zeta^2-1}{\omega_0^2}$$

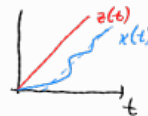
$$A = -C = \frac{2\zeta}{\omega_0^3}$$

$$X(s) = k \left(\frac{1}{s^2} - \frac{2\zeta}{\omega_0^3} \frac{1}{s} + \frac{2\zeta}{\omega_0^3} s + \frac{4\zeta^2-1}{\omega_0^2} \frac{1}{s^2} \right)$$

$$\frac{2\zeta}{\omega_0^3} s + \frac{4\zeta^2-1}{\omega_0^2} = \frac{2\zeta}{\omega_0^3} (s+2\zeta\omega_0 - \frac{\omega_0}{2\zeta}) = \frac{2\zeta}{\omega_0} \left(\frac{s+\zeta\omega_0}{-11-} + \left(\zeta - \frac{1}{2\zeta} \right) \frac{\omega_0\sqrt{1-\zeta^2}}{-11-\sqrt{1-\zeta^2}} \right)$$

$$\downarrow \cos(\omega_0 t) e^{\omega_0 t} \quad \frac{\zeta - \frac{1}{2\zeta}}{11-\zeta^2} \downarrow \sin(\omega_0 t) e^{\omega_0 t}$$

$$x(t) = k \left(t - 2\frac{\zeta}{\omega_0} + \frac{2\zeta}{\omega_0} \left(\cos\omega_0 t + \frac{\zeta - \frac{1}{2\zeta}}{\sqrt{1-\zeta^2}} \sin\omega_0 t \right) e^{-\zeta\omega_0 t} \right)$$



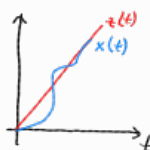
$$H(s) = \frac{\omega_0^2 + 2\zeta\omega_0s}{s^2+2\zeta\omega_0s+\omega_0^2} = 1 - \frac{s^2}{s^2+2\zeta\omega_0s+\omega_0^2}$$

$$z(t) = kt$$

$$Z(s) = \frac{k}{s^2}$$

$$X(s) = \frac{k}{s^2} - \frac{k}{-11-}$$

$$x(t) = kt - \frac{k}{\omega_0\sqrt{1-\zeta^2}} \sin\omega_0 t e^{-\zeta\omega_0 t}$$



$$z(t) = at^2$$

$$Z(s) = \frac{2a}{s^3}$$

$$X(s) = \frac{2a}{s^3} \frac{\omega_0^2 + 2\zeta\omega_0 s}{s^2 + 2\zeta\omega_0 s + \omega_0^2} = 2a \left(\frac{As+B}{I} + \frac{Cs^2+Ds+E}{s^3} \right)$$

$$\omega_0^2 + 2\zeta\omega_0 s = s^2(A+C) + s^2(B+D+2\zeta\omega_0 C) + s^2(E+2\zeta\omega_0 D + \omega_0^2 C) + s(2\zeta\omega_0 E + D\omega_0^2) + E\omega_0^2$$

$$E=1$$

$$D=0$$

$$C = -\frac{1}{\omega_0^2}$$

$$A = \frac{1}{\omega_0^2}$$

$$B = \frac{2\zeta}{\omega_0}$$

$$X(s) = \frac{2a}{s^3} - \frac{2a}{\omega_0^2} \frac{1}{s} + 2a \left(\frac{\frac{1}{\omega_0^2}s + \frac{2\zeta}{\omega_0}}{I} \right)$$

$$\downarrow \quad \downarrow \quad \downarrow$$

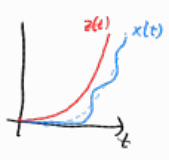
$$at^2 \quad -\frac{2a}{\omega_0^2} \quad \frac{2a}{\omega_0^2} \frac{s+2\zeta\omega_0}{I} = \frac{2a}{\omega_0^2} \left(\frac{s+\zeta\omega_0}{I} + \frac{\zeta}{\sqrt{1-\zeta^2}} \frac{\omega_0 \sqrt{1-\zeta^2}}{I} \right)$$

$$\downarrow \quad \downarrow$$

$$\cos \omega_0 t e^{-\zeta \omega_0 t} \quad -\sin \omega_0 t e^{-\zeta \omega_0 t}$$

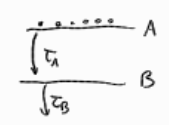
$$x(t) = at^2 - \frac{2a}{\omega_0^2} + \frac{2a}{\omega_0^2} \left(\cos \omega_0 t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_0 t \right) e^{-\zeta \omega_0 t}$$

po dolgem času:
 $\Delta(t) = \frac{2a}{\omega_0^2} = \text{konst.} \rightarrow \text{senzor sledi vhodni z zamikom}$



Uporaba Laplacea za reševanje sistema dif. en.

-dvojnijski sistem



na začetku: $N_A(0) = N_0$; $\dot{N}_A(0)$
 $N_B(0) = 0$

- a) Napiši Kalmanov filter za $\hat{x} = \begin{pmatrix} N_A \\ N_B \end{pmatrix}$
- b) kdaj je korelacija med nivojema največja?

1) dinamika

$$\dot{N}_A = -\frac{1}{\tau_A} N_A$$

$$\dot{N}_B = \frac{1}{\tau_A} N_A - \frac{1}{\tau_B} N_B$$

$$\dot{\hat{x}} = A\hat{x} + \tilde{z} + \Gamma\tilde{w}$$

$$\hat{x} = \begin{pmatrix} \frac{1}{\tau_A} & 0 \\ 0 & -\frac{1}{\tau_B} \end{pmatrix} \hat{x} = \underbrace{\frac{1}{\tau_A} \begin{pmatrix} -1 & 0 \\ 1 & \tau_A/\tau_B \end{pmatrix}}_A \hat{x}$$

$$\tau_A/\tau_B = \tilde{\tau}$$

$$\dot{P} = AP - PA^T$$

$$AP = \frac{1}{\tau_A} \begin{pmatrix} -p_{11} & -p_{12} \\ p_{11} - \tilde{\tau} p_{12} & p_{12} - \tilde{\tau} p_{22} \end{pmatrix}$$

$$\dot{P} = \frac{1}{\tau_A} \begin{pmatrix} -2p_{11} & p_{11} - (\tilde{\tau}+1)p_{12} \\ p_{11} - (\tilde{\tau}+1)p_{12} & 2(p_{12} - \tilde{\tau} p_{22}) \end{pmatrix}$$

$$\dot{p}_{12} = \frac{1}{\tau_A} (p_{11} - (\tilde{\tau}+1)p_{12})$$

$$\dot{p}_{11} = -\frac{2}{\tau_A} p_{11} \quad / \mathcal{L}$$

$$s p_{11}(s) - p_{11}(0) = -\frac{2}{\tau_A} p_{11}(s)$$

$$p_{11}(s) \left(s + \frac{2}{\tau_A} \right) = p_{11}(0)$$

$$p_{11}(s) = \frac{p_{11}(0)}{s + \frac{2}{\tau_A}} \rightarrow p_{11}(t) = p_{11}(0) e^{-\frac{2t}{\tau_A}}$$

$$\dot{p}_{12} = p_{11} - (\tilde{\tau}+1)p_{12} \quad / \mathcal{L}$$

$$s p_{12}(s) - p_{12}(0) = \frac{1}{\tau_A} (p_{11}(s) - (\tilde{\tau}+1)p_{12}(s))$$

$$p_{12}(s) \left(s + \frac{\tilde{\tau}+1}{\tau_A} \right) = \frac{1}{\tau_A} p_{11}(s) = \frac{1}{\tau_A} \frac{p_{11}(0)}{s + \frac{2}{\tau_A}}$$

$$p_{12}(s) = \frac{p_{11}(0)}{\tau_A} \frac{1}{s + \frac{2}{\tau_A}} \frac{1}{s + \frac{\tilde{\tau}+1}{\tau_A}} = \frac{p_{11}(0)}{\tau_A} \left(\frac{\alpha}{s + \frac{2}{\tau_A}} + \frac{\beta}{s + \frac{\tilde{\tau}+1}{\tau_A}} \right)$$

$$\frac{1}{\tilde{\tau}}$$

$$1 = \alpha s + \frac{\beta}{\tilde{\tau}} + \beta s + \frac{2\beta}{\tau_A}$$

$$\alpha = -\beta$$

$$\alpha = \left(\frac{1}{\tilde{\tau}} - \frac{2}{\tau_A} \right)^{-1} = \left(\frac{1}{\tau_A} + \frac{1}{\tau_B} - \frac{2}{\tau_A} \right)^{-1} = \frac{\tau_A \tau_B}{\tau_B - \tau_A}$$

$$p_{12}(s) = \frac{p_{11}(0)}{\tau_A} \frac{\tau_A \tau_B}{\tau_A - \tau_B} \left(\frac{1}{s + \frac{2}{\tau_A}} - \frac{1}{s + \frac{1}{\tilde{\tau}}} \right)$$

$$p_{12}(t) = p_{11}(0) \frac{\tau_B}{\tau_A - \tau_B} \left(e^{-\frac{2}{\tau_A} t} - e^{-\frac{1}{\tilde{\tau}} t} \right)$$

$$\dot{p}_{12}(t) = 0$$

$$\dot{p}_{12} = \frac{p_{11}}{\tau_A} - \frac{\tilde{\tau}+1}{\tau_A} p_{12} = 0$$

$$p_{11}(t) = (\tilde{\tau}+1) p_{12}(t)$$

$$p_{11}(0) e^{-\frac{2t}{\tau_A}} = (\tilde{\tau}+1) p_{12}(0) \frac{\tau_B}{\tau_A - \tau_B} \left(e^{-\frac{2}{\tau_A} t} - e^{-\frac{1}{\tilde{\tau}} t} \right) = (\tilde{\tau}+1) \frac{\tau_B}{\tau_A - \tau_B} e^{-\frac{1}{\tilde{\tau}} t} \left(e^{-\frac{2}{\tau_A} t} - e^{-\frac{1}{\tilde{\tau}} t} \right)$$

$$e^{-\frac{2}{\tau_A} t} \left(1 - \frac{\tau_A \tau_B}{\tau_A - \tau_B} \left(1 - e^{-\frac{1}{\tilde{\tau}} t} e^{-\frac{2}{\tau_A} t} \right) \right) = 0$$

$$\frac{\tau_A - \tau_B}{\tau_A + \tau_B} = 1 - e^{-\frac{2}{\tau_A} t} e^{-\frac{1}{\tilde{\tau}} t} \quad / \ln$$

$$-t \left(\frac{1}{\tau_A} - \frac{1}{\tau_B} \right) = \ln \frac{2\tau_B}{\tau_A + \tau_B}$$

$$t_m = \frac{\tau_A \tau_B}{\tau_A + \tau_B} \ln \frac{\tau_A + \tau_B}{2\tau_B}$$